Data privacy

List of exercises

Concepts

Exercise 1. Discuss the transparency principle in data privacy in terms of risk and utility.

Exercise 2. Justify the importance of the transparency principle in terms of attacks and risk.

Exercise 3. Why anonymity sets are of relevance for privacy? Does an anonymity set protect against identity disclosure? Does an anonymity set protect against attribute disclosure? Why?

Disclosure and privacy models

Exercise 4. Define identity disclosure for a database with information about people. Define attribute disclosure.

Exercise 5. Why the application of univariate microaggregation to a database with several variables does not necessarily produce a file that satisfies k-anonymity?

Exercise 6. How should variables be selected for microaggregation so that k-anonymity is satisfied?

Exercise 7. Relate k-anonymity, as a privacy model, with its parameter k and anonymity sets.

Exercise 8. Give an example of disclosure from a file that satisfies k-anonymity. Give an example of a privacy model that provides better privacy guarantees against such type of disclosure.

Exercise 9. What is the point of l-diversity if k-anonymity already protects against identity disclosure?

Exercise 10. Define briefly the following privacy models, mention if they focus on a database release or on a query release, and if they focus on attribute disclosure or on identity disclosure.

- Re-identification
- Uniqueness
- Result-driven
- k-Anonymity
- k-Confusion

- p-Sensitive k-anonymity
- k-Anonymity l-diversity
- k-Anonymity t-closeness
- Interval disclosure
- Differential privacy
- Local differential privacy
- Integral privacy
- Homomorphic encryption
- Secure multiparty computation

Exercise 11. In what sense the privacy guarantees of k-confusion are the same as the privacy guarantees of k-anonymity.

Exercise 12. Why k-confusion provides a similar protection than k-anonymity if records can be different? How the definition of k-confusion links with the concept of anonymity set.

Exercise 13. Discuss whether the following statement is always true or not, and its connection with k-confusion: "Records need to be indistinguishable in order to build an anonymity set".

Exercise 14. Explain what a membership inference attack is.

Exercise 15. GANs are used to generate synthetic data. Does it make sense to apply membership inference attacks to GANs? Why?

Privacy for data: Masking methods

Exercise 16. Discuss whether the following statement is true or false: Any masking method that includes some randomization is not vulnerable to transparency attacks.

Exercise 17. In microaggregation we require clusters to have at least k records, and it is usual that the maximum size is $2k$. What is the role of having a maximum size?

Exercise 18. Let us consider the application of PRAM to a categorical dataset with a transition matrix that has some zeros in non-diagonal positions. Can this matrix provide a meaningful differential privacy guarantee of the protected file?

Exercise 19. What is PRAM and what is the connection between PRAM and differential privacy?

Exercise 20. What condition needs the transition matrix in PRAM to provide meaningful differential privacy?

Exercise 21. Why just synthetic data (using e.g. a type of GAN) is not enough? Why do we need, for example, differentially private synthetic data?

Exercise 22. We need to apply a masking method to a database to produce a database with low disclosure risk. We consider different parameterizations for this method. For example, consider the case of applying microaggregation with different cluster sizes. How are we going to select the best parameterization?

Exercise 23. How heuristic methods for microaggregation work? Explain its relation with a privacy model.

Privacy for computations: Differential privacy mechanisms

Exercise 24. You and me want to know in our wedding whether your guests or my guests have larger income. We ask them to provide us with their salary using local differential privacy.

All guests send us their salary with a noise following the distribution $L(0, \Delta s/\epsilon)$, with appropriate sensitivity and ϵ. Then we add the values from your guests, and the values from my guests, and compare them.

Discuss: Does this allows us to know if the average salary of your guests is larger than the average salary of my guests?

Exercise 25. We are a group of thieves, and a painting has just been stolen from an important museum. We want to know if any of us was involved, but without revealing who it was. Can a Secure Multiparty Computation protocol help us find out if one of us was responsible, without anyone learning who it was? Discuss the answer. (Assume there are at least three of us.)

Exercise 26. Let D represent a data set. Then, let us consider the application of a machine learning algorithm to D to obtain a machine learning model M. Explain why M can lead to disclosure, and give an example.

Exercise 27. In differential privacy, why multiple queries on the same dataset increase the disclosure. Is this quantifiable in terms of the epsilon value?

Exercise 28. Let us consider the application of differential privacy to the following functions, we assume that the values to be combined are in the interval [0,1]

- (a) minimum of an arbitrary number of arguments.
- (b) maximum of an arbitrary number of arguments.
- (c) median of an arbitrary number of arguments. Recall that for an odd number of arguments, the median is the value that occupies the central position after sorting all arguments. For an even number of arguments, the median is the average of the two values in central positions.

Questions

- Compute the sensitivity of these functions. Compute the Laplace scale parameter b when we require $\epsilon = 1$ and $\epsilon = 2$.
- What will provide a more accurate output?

– Compare the result with the one of the mean.

Exercise 29. Let us consider that we want to compute the function f from a database X. Let this function be the mean of a variable. So, we know that the sensitivity of f is $(max - min)/S$ where S is the minimal number of records in S.

Let us consider that we first microaggregate X with at least k records in each cluster.

Can we say something about the sensitivity of the function f? Is it larger or smaller?

Exercise 30. Let us consider ϵ-differential privacy implemented using the Laplacian mechanism. Describe the general effect of sensitivity of a query on the amount of noise you would add, assuming a fix ϵ.

Exercise 31. Let us consider two functions, one that provides a differentially private mean and another that provides a differentially private standard deviation. Then, let us consider a data set. If we apply both functions, each with a privacy budget ϵ to this set, what can we say about the disclosure?

Exercise 32. A data scientist wants to apply differential privacy to a data set. First, the data scientist queries the data set to have the maximum and the minimum of the data set. Then, these two values are used to compute the sensitivity. Finally, the data scientist applies local differential privacy using this sensitivity and the desired ϵ value. What is wrong with this method from a data privacy perspective?

Exercise 33. Let us consider data in [0,1000] and with a database with at least 100 records. Consider the following two functions mean and maximum applied to this database, in the context of differential privacy assuming a preselected ϵ.

Which of these functions requires more noise, why? Discuss the noise in relation to function sensitivity.

Other privacy mechanisms

Exercise 34. We can consider Secure Multiparty Computation (SMC) and differential privacy (DP) as two different privacy models. Why should we consider them together in a single framework? Give an example.

Exercise 35. Why the problem of dining cryptographers (or, equivalently, computation of exclusive or using secure multiparty computation) can be seen as related to unobservability?

Exercise 36. What is a semi-honest adversary?

Exercise 37. What is a malicious adversary? and why malicious adversaries can get some information from a secure multiparty computation protocol if all the information is encrypted?

Exercise 38. What is the difference between a semi-honest adversary and a malicious adversary in secure multiparty computation?

Exercise 39. Why in tabular data protection, we need secondary suppressions? Why the suppression of sensitive values is not enough?

Exercise 40. Perturbative methods appear repeatedly in data protection, for different types of data. List some of them.

Selection of data protection mechanisms

Exercise 41. Consider a distributed scenario as in federated learning where you have your own data in your own device and you fully ignore the data of others. Discuss why local anonymization (e.g. k-anonymity or local differential privacy) may not be enough for guaranteeing your privacy.

Exercise 42. How information loss for a data usage f can be defined?

Exercise 43. What is the difference between generic and specific information loss measures?